

MONTPELLIER MEDITERRANEE AP, France

WMO: 076430

Lat: 43.5769N

Lon: 3.9631E

Elev: 8

StdP: 101.23

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -2.0	(c) -0.4	(d) -11.1	(e) 1.5	(f) 2.5	(g) -8.6	(h) 1.8	(i) 3.7	(j) 15.0	(k) 11.7	(l) 13.1	(m) 10.6	(n) 2.7	(o) 10	(p) 0.499

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 9.7	(c) 32.2	(d) 20.4	(e) 30.9	(f) 20.2	(g) 29.6	(h) 19.9	(i) 23.9	(j) 27.6	(k) 23.0	(l) 26.9	(m) 22.2	(n) 26.3	(o) 5.2	(p) 200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
22.8	17.5	26.0	21.8	16.5	25.2	20.9	15.6	24.5	71.5	27.5	68.2	26.9	65.2	26.3	26.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12.4	10.8	9.4	-4.6	35.2	1.8	1.8	-6.0	36.6	-7.0	37.6	-8.1	38.7	-9.4	40.0
			-5.8	24.9	2.0	0.8	-7.3	25.5	-8.5	25.9	-9.6	26.4	-11.1	27.0
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	15.4	7.6	8.2	11.3	13.8	17.5	21.8	24.3	23.9	20.1	16.5	11.4	8.2
	DBStd	6.56	3.22	3.18	2.81	2.60	2.68	2.83	2.36	2.16	2.46	2.93	3.33	3.34
	HDD10.0	269	86	64	18	2	0	0	0	0	0	1	23	74
	HDD18.3	1623	332	284	219	137	46	4	1	1	10	69	208	314
	CDD10.0	2252	13	14	58	117	234	354	443	431	303	203	65	18
	CDD18.3	565	0	0	0	2	22	108	186	173	63	13	0	0
	CDH23.3	4381	0	0	3	14	139	826	1619	1397	344	37	1	0
	CDH26.7	1284	0	0	0	1	24	237	541	417	59	5	0	0
Wind	WSAvg	4.2	4.1	4.3	4.5	4.7	4.4	4.1	4.2	4.0	3.9	4.2	4.3	4.0
Precipitation	PrecAvg	869	87	76	77	60	49	46	22	40	103	123	77	79
	PrecMax	1374	245	249	279	148	128	147	61	99	293	298	260	244
	PrecMin	497	6	1	0	1	6	2	1	1	17	13	19	4
	PrecStd	276	69	71	77	45	36	40	15	32	80	84	60	58
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	18.1	18.8	23.0	25.1	29.3	33.7	34.0	33.5	30.4	26.6	21.0	17.7
		MCWB	12.3	11.7	13.7	15.5	18.9	21.3	20.7	21.3	19.0	17.2	15.3	12.7
	2%	DB	15.7	16.5	20.0	22.9	26.6	31.2	32.3	31.7	28.2	24.1	18.9	15.9
		MCWB	10.9	10.5	12.2	14.4	17.4	20.4	20.3	20.5	18.4	17.5	15.2	12.4
	5%	DB	14.0	14.8	18.2	21.0	24.7	29.3	31.0	30.3	26.8	22.6	17.8	14.8
		MCWB	10.2	9.8	11.5	13.6	16.4	19.6	20.3	20.2	18.3	17.5	14.5	11.9
	10%	DB	12.9	13.5	16.5	19.2	23.1	27.6	29.7	29.1	25.2	21.3	16.8	13.6
		MCWB	9.9	9.6	10.8	12.7	15.6	18.9	20.0	20.1	17.9	17.2	13.8	10.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	13.1	12.7	14.6	17.4	20.3	23.7	25.0	25.2	22.5	21.1	17.7	14.6
		MCDB	16.2	16.3	19.9	22.2	26.4	29.0	28.8	28.4	25.1	22.6	18.9	16.1
	2%	WB	12.1	11.9	13.4	15.9	18.9	22.5	23.8	24.1	21.7	20.0	16.5	13.5
		MCDB	14.3	14.5	17.5	20.4	24.0	27.5	27.8	27.5	24.3	21.7	18.1	15.1
	5%	WB	11.3	11.2	12.7	14.8	17.9	21.5	22.8	23.3	20.8	19.0	15.4	12.3
		MCDB	13.3	13.7	16.2	18.7	22.3	26.6	27.2	26.7	23.8	21.3	17.2	14.0
	10%	WB	10.3	10.4	12.1	14.0	17.1	20.6	22.0	22.5	20.0	18.1	14.4	11.3
		MCDB	12.4	13.1	15.3	17.2	21.0	25.5	26.4	26.1	23.5	20.3	16.3	13.1
Mean Daily Temperature Range	5% DB	MDBR	8.2	9.1	9.6	9.2	9.3	9.7	9.7	9.7	9.5	8.4	8.3	8.1
		MCDBR	9.3	11.0	12.1	11.9	11.9	11.7	11.4	11.4	11.4	10.0	8.7	8.6
		MCWBR	6.2	6.9	6.7	6.2	5.6	5.2	4.8	5.1	5.3	5.4	5.7	5.9
	5% WB	MCDBR	7.5	9.1	9.7	9.8	9.8	9.6	9.0	8.6	8.3	6.9	6.4	6.9
		MCWBR	5.7	6.3	6.1	6.0	5.5	5.1	5.1	5.2	5.3	4.8	5.0	5.3
Clear-Sky Solar Irradiance	taub	0.323	0.340	0.379	0.404	0.412	0.414	0.409	0.407	0.392	0.387	0.349	0.320	
	taud	2.479	2.430	2.337	2.285	2.314	2.341	2.355	2.365	2.393	2.406	2.461	2.495	
	Ebn at Noon	806	852	856	861	862	859	859	848	833	785	765	774	
	Edh at Noon	72	89	111	126	126	123	120	115	104	91	74	66	
All-Sky Solar Radiation	RadAvg	1.70	2.67	3.93	5.15	6.21	7.19	7.19	6.13	4.66	2.89	1.89	1.48	
	RadStd	0.19	0.26	0.31	0.41	0.42	0.34	0.30	0.27	0.27	0.30	0.23	0.16	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+39
(47) Regional (32 neighbors)		N/A	N/A	N/A	+0.71	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page